

Trainings ▾

Preparatory Steps

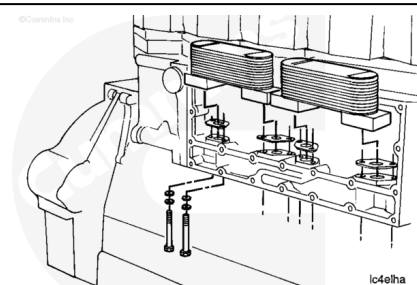
- Remove the oil cooler cover. Refer to Procedure 007-045 in Section 7 (</qs3/pubsys2/xml/en/procedures/18/18-007-045-tr.html>).

Remove

There is little clearance between the capscrews and the block on some engines. If necessary, use a drift and a mallet to tap the loosened capscrews from the oil cooler housing.

Remove the eight capscrews and the two oil cooler elements.

Remove and discard the gaskets.



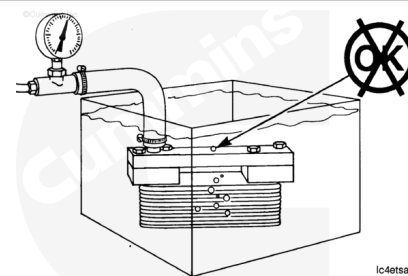
Pressure Test

Note : Cummins, Inc. does **not** recommend any form of cleaning of oil cooler elements (internal or external). This includes cleaning with solvent, flushing with oil, steam cleaning, and ultrasonic cleaning.

If the engine has experienced a malfunction that allows debris to pass through the lubrication system to the full-flow oil filters and oil passages resulting in the lubricating oil cooler become contaminated, the oil cooler elements **must** be replaced.

Inspect the oil cooler elements externally for scale build-up, corrosion, embedded debris, and/or any damage to the elements. If there is any evidence of any of these conditions, the oil elements **must** be replaced.

Note : Contamination can be defined by metal, rubber, or other solid particles present in the lubricating oil



filter and/or by debris damage to the bearings, or debris damage of any other components such as they do **not** meet reuse guidelines. Remove gasket material from the oil cooler cover, oil cooler element, and the engine block mating surface.

Heating the water in the tank to 50°C [120°F] will improve the test results.

Pressure check the elements to check for leaks.

Measurements		
	kpa	psi
Air Pressure	415	60

If leaks are found, the elements **must** be replaced.

Install

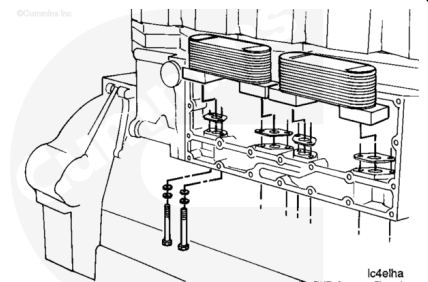
Note : Remove the plastic shipping plugs from the new oil cooler elements.

In early 1988 the threads in the oil cooler element mounting feet changed. The older elements had fine threads (24 threads per inch). The new elements have coarse threads (16 threads per inch). Service oil cooler elements are supplied with capscrews that have coarse threads. Make sure the proper capscrews are used or the oil cooler element will be damaged beyond reuse.

Install the gaskets, oil cooler elements, and capscrews.

Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



Finishing Steps

- Install the oil cooler cover. Refer to Procedure 007-045 in Section 7 (</qs3/pubsys2/xml/en/procedures/18/18-007-045-tr.html>).

